

01

Introduction to Offensive & Defensive Technologies

Kamilimu cohort 10 - cybersecurity



How would you define cyber security?

What is cybersecurity? Why does it matter?

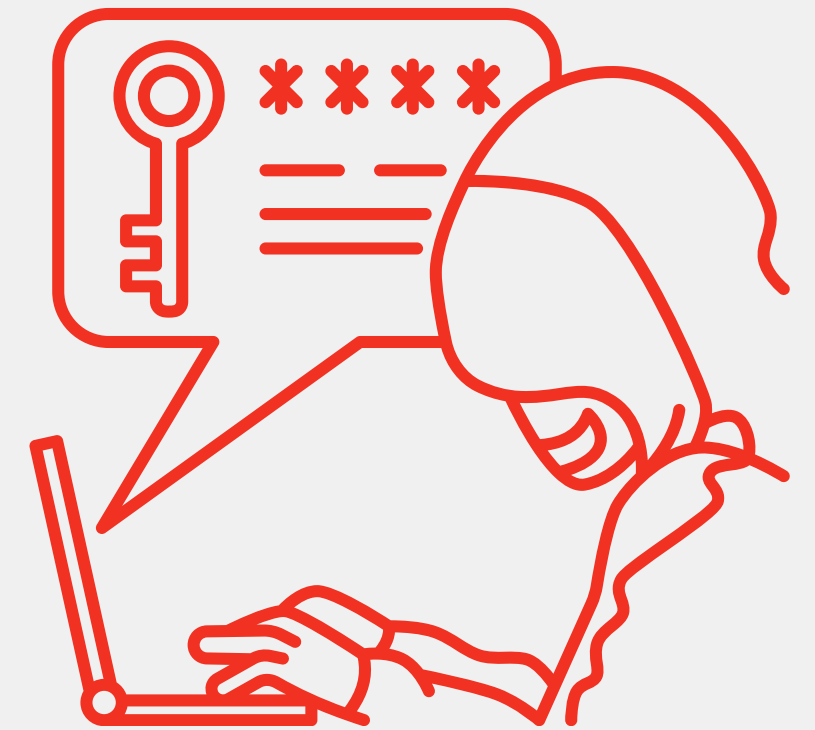
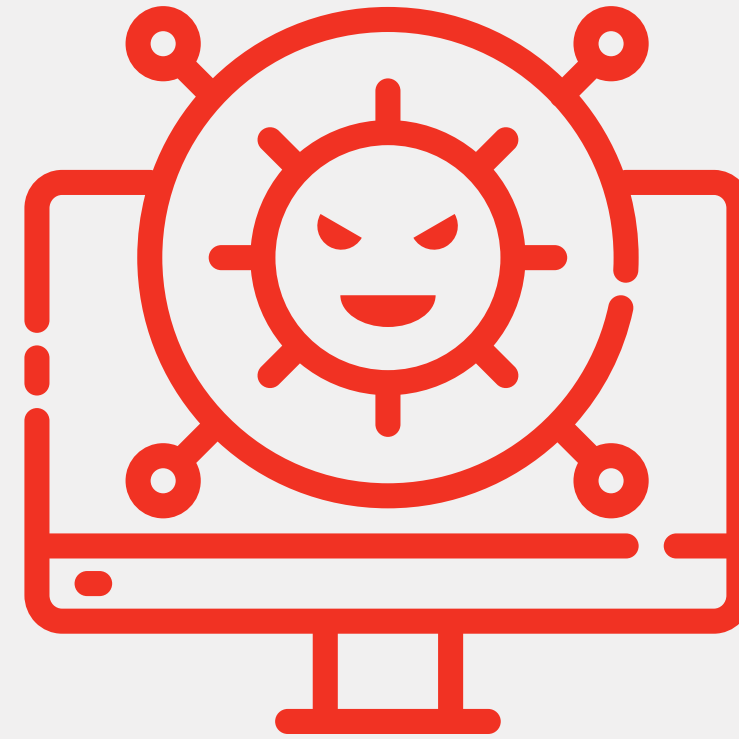


02

- Who does cybersecurity impact?
- If cybersecurity did not exist, would it matter?

A few real world cyber events

1. Do you think any of these are possible here in Africa, Kenya?
2. Which ones surprised you?
3. What do you think the impact of some of these attacks were?
4. How could they have been stopped/prevented?



Core Principles of cybersecurity (C.I.A triad)



Confidentiality

Is my data or my technology (e.g. servers) only accessible to those who should see it?



Integrity

Does the data I have remain unmodified within the systems it resides in?



Availability

Am I able to access what i need to when I need to without undue disruption?



How is cybersecurity practiced?



Offensive security

The aspects of security that either mimic or actively work to harm systems, applications, and networks. i.e. Red Team/ Black hat, White hat, Grey Hat.

- Can involve a wide range of actions including malware development & deployment, hacktivism, penetration testing and vulnerability research etc.
- Unfortunately can be performed by both good and bad actors for a wide variety of reasons.

Defensive Security

The aspects of security that work to protect systems, applications and networks i.e. Blue Team.

- Can span a wide range of possibilities including tools that protect such as Firewalls and Cryptography to people who protect
- Interesting aspects of defensive security include:
 - Active monitoring: Security Operations Centres (SOC), Threat Modelling, Intelligence and Architecture.

What are the varying aspects of cybersecurity?



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Digital Forensics & Incident Response (DFIR)

The aspect of security that works to find out what happened when things go wrong. The investigative arm of security.
Think, CSI

Blended security - merging two or more aspects of security

- Purple team: merging both Red (Offense) and Blue (Defense) work

Governance, Risk and Compliance (GRC)

Where analysts think about how laws, international frameworks and regulation impact an entity's security.

- For example, in US, healthcare data is governed by HIPAA and in Kenya, the CyberCrime Act and Data Protection Act/Regulations play a huge role in organizations security posture

Imagine a Bank...



Confidentiality

What would a bank need to keep confidential?



Integrity

What would a bank need to ensure does not change no matter what?



Availability

What would a bank need to make sure stays available at all times?



How would you break, protect, fix (solve) the bank's security?

01

Offense

- Reconnaissance: web, nmap, phishing
- Exploitation: hydra, burpsuite, netexec
- Post Exploitation: Cloacking, AMSI bypass etc.

02

Defense

- Detection: SIEM, Wireshark, EDR/XDR
- Analysis and Containment: ACL, EDR/XDR, IPS
- Treatment, recovery and restoration: Backup and Restoration tooling

03

Forensics

- Preparation: scoping tools
- Detection and Analysis: Wireshark, VirusTotal
- Containment, Eradication & Recovery: Imaging tools
- Post-Incident activities:



Resources

Offensive Security

- Reading: <https://www.ibm.com/think/topics/offensive-security>
- Resource: <https://www.netacad.com/courses/ethical-hacker?courseLang=en-US> (Free Certification)

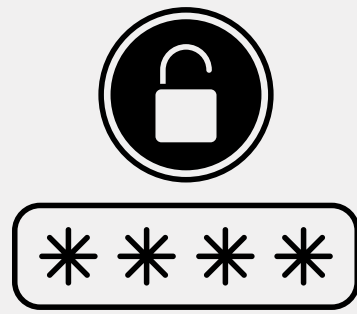
Defensive Security

- Reading & Resource: <https://www.netacad.com/career-paths/splunk-cybersecurity-defense-analyst?courseLang=en-US>

Digital Forensics (DFIR)

- Reading: <https://www.bluevoyant.com/knowledge-center/understanding-digital-forensics-process-techniques-and-tools>
- Resource: <https://dfirdiva.com/> & <https://www.dfir.training/>





Questions?

